

- 2 (a) (i) (vertical component = $44 \sin 30^\circ = 22$ N) A1
- (ii) (horizontal component = $44 \cos 30^\circ = 38(.1)$ N) A1
- (b) $W \times 0.64 = 22 \times 1.60$ C1
(W =) 55 N A1
- (c) For a system in equilibrium, net force = 0 B1
- F has a horizontal component (not balanced by W) B1
or F has 38 N acting horizontally
or 38 N acts on wall
or vertical component of F does not balance W
or F and W do not make a closed triangle of forces
- (d) line from P towards point on wire vertically above W and direction up B1
three non-parallel coplanar forces must act through the same line B1
- (e) line from P towards right B1
line from P towards point on wire vertically below W and direction down B1

3 (a) (i) Any difference about Amplitude/Phase difference/Energy R1